PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review
AUTHORS	Nnaji, Chukwudi; Kuodi, Paul; Walter, Fiona; Moodley, Jennifer

VERSION 1 – REVIEW

REVIEWER	Lydia Pace Brigham and Women's Hospital, Boston, Massachusetts, Medicine
REVIEW RETURNED	18-Oct-2021

GENERAL COMMENTS	Thanks for the opportunity to review this manuscript, which makes an important contribution to the literature by reviewing the published studies of breast and cervical cancer early diagnosis interventions targeting symptomatic individuals in low- and middle-income countries. Overall, the study question, methods and results are clearly articulated. I have some concerns about the study inclusion process and quite a few suggestions to strengthen the discussion. Introduction: -Would clarify a bit more in the introduction (eg through one additional sentence) that you were focused on reducing delays among individuals with symptoms –I think this is important to be very clear about this up front particularly since the global cervical cancer literature in particular is so focused on screening. For example you can state that given the evidence demonstrating that diagnostic delays among symptomatic women are so long in LMICs, interventions to reduce these delays among people with symptoms would likely have substantial impact on outcomes and are also a prerequisite for screening programs. It is not clear until the methods section that you excluded screening studies. Methods -The use of the WHO Guide to Early Cancer Diagnosis as a framework and to divide up the studies is a strength. The search strategy and terms seem reasonable overall. -Although the authors only included studies that “aimed to influence the timeliness of diagnosis”, they don’t seem to have actually required studies to examine timeliness (for example, they included some that just measured provider skills). I think it would have been much better to only include studies that described outcomes related to timeliness, since the lack of such outcomes in some of the included studies seems to add some subjectivity to the search strategy and results, and I do see some inconsistencies with regard to studies that are included/ excluded which may result from this. From this manuscript it is not clear how the authors decided whether a study was aimed at increasing timeliness. Results Figures and tables are well-selected and clear. However, in Table 1 I

	do think it would be valuable to include the primary outcome examined for each study, since the limitations of the outcomes examined are a major focus of the discussion. I think that the authors may be overusing the term “quasi-experimental.” I believe that a simple pre/post design is not considered to be quasi-experimental. (Table 1 and text of results) Overall, the language used to describe the study designs seems a bit imprecise (e.g. instead of “trial” say “randomized clinical trial”; instead of “retrospective study” would say “retrospective cohort study” if appropriate). Discussion Overall, the discussion needs some work. The authors tend to dwell in generalities and don’t make specific recommendations for outcomes, study designs, and effective interventions. They could provide a much more cogent summary of the state of the literature and what kinds of studies are needed. The authors discuss limitations of the included articles’ outcome measures, but they do this in general terms and do not outline specifically what measures would actually be most clinically relevant and meaningful and how these could be standardized. I think it would be worth articulating that the gold standard outcome for an early diagnosis program would be reduced breast cancer mortality, which would require an appropriate control group and a very long follow period. Articulating what the authors consider to be clinically meaningful proxies (for example, downstaging, which I feel would be more valuable than time to diagnosis) would be helpful and would also allow the authors to make a stronger statement about gaps in the literature. There is also, of course, a role for studies that show improvements in provider skills and other process measures. Some of the methodological challenges with assessing time to diagnosis (e.g. recall bias, and differing definitions across settings) may be worth mentioning as well. How could time to diagnosis measurements be “standardized,” as the authors suggest in their concluding paragraph? Similarly, discussing more specifically the quality limitations of the existing study designs and the most rigorous study designs would be helpful – all sentences about study quality in the discussion are very general. It seems worth discussing the limited studies on symptomatic cervical cancers. I am not sure whether this dearth reflects the appropriate degree of focus on cervical cancer screening, for which there is strong evidence (and perhaps women with cervical cancer symptoms are in fact evaluated in such programs), the fact that when cervical cancers are symptomatic, they are likely to be at an advanced stage, or stigma/ shame around cervical cancer symptoms. But this does raise the question of whether cervical cancer should be included in this review. In either case, the limited representation of cervical cancer here deserves more mention. The second and third paragraphs on page 21 (one about intervention challenges, one about Covid) are weak – they are too wordy, and don’t contribute to the overall flow of the discussion. They are also too general. If the authors feel that the reviewed articles describe important barriers/ challenges to early diagnosis interventions, this should be described in more detail. However, since this wasn’t an outcome examined in the review, I’d consider removing. The Covid paragraph feels quite tangential as it is written – what ARE the implications of these studies for mitigating Covid-related delays? The limitations section seems to have a number of typos/ unclear phrases, eg:

	-“Thirdly, the short duration of the majority of the interventions undermines poses additional limitations to the applicability of the evidence.” Not clear. -Line 48: what does “comprehensive enough” mean? -Last sentence of the paragraph – this review excluded screening studies, so it explicitly does NOT assess the effectiveness of screening. However, it’s possible that some screening studies included early diagnosis components (e.g. included women with symptoms), and these are not assessed.

REVIEWER	Smita Asthana Institute of Cytology and Preventive Oncology, Indian Council of Medical Research, Noida, India, epidemiology and biostatstics
REVIEW RETURNED	07-Jan-2022

GENERAL COMMENTS	This is a well written paper to analyse evidences on nature and effectiveness of of intervention for timely diagnosis of breast and cervical cancer in low and middle income countries. The topic is interesting and important and focusing on two major women related cancer the importance of topic increased as it is about timely diagnosis of these cancers in low and middle income countries. The paper clearly demarcate and select the primary papers related to the topic of interventions and there is no confusion and mixing of other aspect such as screening related research. This is an important paper and needs to be publish.

REVIEWER	Nicole Campos Harvard University T H Chan School of Public Health, Health Policy and Management
REVIEW RETURNED	13-Jan-2022

GENERAL COMMENTS	“Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review” is timely, given the disruptions to global health care services during the COVID-19 pandemic. Rather than addressing interventions to improve access to screening, this systematic review focuses on steps in the diagnostic process including awareness of symptoms/access to care; clinical evaluation, diagnosis, and staging; and access to treatment/palliation, as outlined in the 2020 WHO Report on Cancer. Strengths of the review include the use of PRISMA protocols and the restriction to LMIC settings (based on World Bank classification using GNI per capita) over the past 10 years. Major comments: • The Results are nicely described in Table 1 by geographical region, study design, target population, cancer type, and intervention placement within the WHO essential steps framework. • The authors also present the risk of bias assessment results, which is useful given the low number of studies with low risk of bias. • P.16, line 22: Is it correct that the proportion of DCIS was 23.1% in screening groups but only 2.5% in the diagnostic group? It seems like the cancer detection rates were higher in the diagnostic group a few lines prior. • The impact of Step 1 interventions appears to be minimal and difficult to measure/report across studies (e.g., a 13 day difference in time to diagnosis; a cervical cancer prevention education program for high school students, who are not really at risk for cervical cancer yet due to their young age), yet the authors seem to imply these can

	be effective in the Discussion ("feasible and can yield desirable outcomes"). Can the authors comment on what a clinically relevant difference in time to diagnosis is for breast and cervical cancer? • Results: Most of the studies do not include measures for time to diagnosis. I realize this is one of the points the authors are trying to make – that too few studies consider timeliness. Is there a way to provide a summary / number for timeliness, since it is a main outcome of interest in the paper?

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Comment: Thanks for the opportunity to review this manuscript, which makes an important contribution to the literature by reviewing the published studies of breast and cervical cancer early diagnosis interventions targeting symptomatic individuals in low- and middle-income countries. Overall, the study question, methods and results are clearly articulated. I have some concerns about the study inclusion process and quite a few suggestions to strengthen the discussion.
Response: We are grateful for this acknowledgment of the strength and importance of our review and for the suggestions to improve the discussion.
Comment: Would clarify a bit more in the introduction (eg through one additional sentence) that you were focused on reducing delays among individuals with symptoms –I think this is important to be very clear about this up front particularly since the global cervical cancer literature in particular is so focused on screening. For example you can state that given the evidence demonstrating that diagnostic delays among symptomatic women are so long in LMICs, interventions to reduce these delays among people with symptoms would likely have substantial impact on outcomes and are also a prerequisite for screening programs. It is not clear until the methods section that you excluded screening studies.
Response: We appreciate this insightful suggestion. We have made the clarification as suggested (Lines 119 – 124)
Comment: The use of the WHO Guide to Early Cancer Diagnosis as a framework and to divide up the studies is a strength. The search strategy and terms seem reasonable overall.
Response: Thank you.
Comment: Although the authors only included studies that "aimed to influence the timeliness of diagnosis", they don't seem to have actually required studies to examine timeliness (for example, they included some that just measured provider skills). I think it would have been much better to only include studies that described outcomes related to timeliness, since the lack of such outcomes in some of the included studies seems to add some subjectivity to the search strategy and results, and I do see some inconsistencies with regard to studies that are included/ excluded which may result from this. From this manuscript it is not clear how the authors decided whether a study was aimed at increasing timeliness.
Response: We had clarified that eligible studies had to report outcomes relevant to any of the three steps of the WHO Guide to Cancer Early Diagnosis (See Lines 152 – 155). These would, for instance, include studies assessing interventions seeking to reduce access delays by promoting awareness, help-seeking and linkage to diagnostic services (Step 1), which report outcomes such as improved awareness and help-seeking as proxies of diagnostic timeliness; and may or may not explicitly report time-related outcomes. We have clarified this further (Lines 160 – 161). We applied these criteria strictly in the study screening and selection process and have acknowledged the limitation of these proxy outcomes (Lines 450 – 455)

Comment: Figures and tables are well-selected and clear. However, in Table 1 I do think it would be valuable to include the primary outcome examined for each study, since the limitations of the outcomes examined are a major focus of the discussion.

Response: Thank you for this suggestion. We have added a new column for primary outcomes in Table 1.
Comment: I think that the authors may be overusing the term “quasi-experimental.” I believe that a simple pre/post design is not considered to be quasi-experimental. (Table 1 and text of results)
Overall, the language used to describe the study designs seems a bit imprecise (e.g. instead of “trial” say “randomized clinical trial”; instead of “retrospective study” would say “retrospective cohort study” if appropriate).
Response: We have described the study designs more precisely (Lines 225 -227, 262, 268, 275, 324, 329 and Table 1)
Comment: Overall, the discussion needs some work. The authors tend to dwell in generalities and don’t make specific recommendations for outcomes, study designs, and effective interventions. They could provide a much more cogent summary of the state of the literature and what kinds of studies are needed.
Response: We have revised the discussion for more coherence (Lines 390 – 392, 404 – 407, 411 – 416, 422 – 459, 463 – 477, 482 - 483).
Comment: The authors discuss limitations of the included articles’ outcome measures, but they do this in general terms and do not outline specifically what measures would actually be most clinically relevant and meaningful and how these could be standardized. I think it would be worth articulating that the gold standard outcome for an early diagnosis program would be reduced breast cancer mortality, which would require an appropriate control group and a very long follow period. Articulating what the authors consider to be clinically meaningful proxies (for example, downstaging, which I feel would be more valuable than time to diagnosis) would be helpful and would also allow the authors to make a stronger statement about gaps in the literature.
Response: We appreciate this insightful feedback. We have revised as suggested (line 446 – 456)
Comment: Some of the methodological challenges with assessing time to diagnosis (e.g. recall bias, and differing definitions across settings) may be worth mentioning as well..
Response: We have taken these limitations into account (Lines 473 – 476)
Comment: How could time to diagnosis measurements be “standardized,” as the authors suggest in their concluding paragraph?
Response: We have clarified this in the discussion (Lines 455-457)
Comment: Similarly, discussing more specifically the quality limitations of the existing study designs and the most rigorous study designs would be helpful – all sentences about study quality in the discussion are very general.
Response: We have taken this into account in the discussion (Lines 449 – 452)
Comment: It seems worth discussing the limited studies on symptomatic cervical cancers. I am not sure whether this dearth reflects the appropriate degree of focus on cervical cancer screening, for which there is strong evidence (and perhaps women with cervical cancer symptoms are in fact evaluated in such programs), the fact that when cervical cancers are symptomatic, they are likely to be at an advanced stage, or stigma/ shame around cervical cancer symptoms. But this does raise the question of whether cervical cancer should be included in this review. In either case, the limited representation of cervical cancer here deserves more mention.
Response: We have addressed this (lines 439 – 445)

Comment: The second and third paragraphs on page 21 (one about intervention challenges, one about Covid) are weak – they are too wordy, and don’t contribute to the overall flow of the discussion. Covid paragraph feels quite tangential as it is written – what ARE the implications of these studies for mitigating Covid-related delays? They are also too general.
Response: We have revised it for clarity and placed more coherently (See lines 467 – 472)

Comment: If the authors feel that the reviewed articles describe important barriers/ challenges to early diagnosis interventions, this should be described in more detail. However, since this wasn’t an outcome examined in the review, I’d consider removing.

Response: We appreciate these thoughtful remarks. We have mentioned only intervention-related factors associated with intervention outcomes where such is reported in the studies (for instance, lines 301 – 302 and 364 - 366)

Comment: The limitations section seems to have a number of typos/ unclear phrases, eg:
-"Thirdly, the short duration of the majority of the interventions undermines poses additional limitations to the applicability of the evidence." Not clear.
Response: We have clarified the statement (Lines 480 - 482) and corrected typos. Thank you.

Comment: Line 48: what does "comprehensive enough" mean?
Response: The statement has been clarified (Lines 482 – 483)

Comment: Last sentence of the paragraph – this review excluded screening studies, so it explicitly does NOT assess the effectiveness of screening. However, it's possible that some screening studies included early diagnosis components (e.g. included women with symptoms), and these are not assessed.
Response: We have revised (lines 483 – 486). Thank you for the thoughtful suggestion.

Reviewer 2
Comment: This is a well written paper to analyse evidences on nature and effectiveness of of intervention for timely diagnosis of breast and cervical cancer in low and middle income countries. The topic is interesting and important and focusing on two major women related cancer the importance of topic increased as it is about timely diagnosis of these cancers in low and middle income countries. The paper clearly demarcates and select the primary papers related to the topic of interventions and there is no confusion and mixing of other aspect such as screening related research. This is an important paper and needs to be published.
Response: We appreciate this encouraging and valuable feedback. Thank you.

Reviewer 3
Comment: "Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review" is timely, given the disruptions to global health care services during the COVID-19 pandemic. Rather than addressing interventions to improve access to screening, this systematic review focuses on steps in the diagnostic process including awareness of symptoms/access to care; clinical evaluation, diagnosis, and staging; and access to treatment/palliation, as outlined in the 2020 WHO Report on Cancer. Strengths of the review include the use of PRISMA protocols and the restriction to LMIC settings (based on World Bank classification using GNI per capita) over the past 10 years.
Response: We are grateful for this acknowledgment of the strength of our review.
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Comment: The Results are nicely described in Table 1 by geographical region, study design, target population, cancer type, and intervention placement within the WHO essential steps framework.
Response: Thank you.
Comment: The authors also present the risk of bias assessment results, which is useful given the low number of studies with low risk of bias.
Response: We appreciate this.

Comment: P.16, line 22: Is it correct that the proportion of DCIS was 23.1% in screening groups but only 2.5% in the diagnostic group? It seems like the cancer detection rates were higher in the diagnostic group a few lines prior.
Response: We have clarified this (Lines 284 – 287)

Comment: The impact of Step 1 interventions appears to be minimal and difficult to measure/report across studies (e.g., a 13 day difference in time to diagnosis; a cervical cancer prevention education program for high school students, who are not really at risk for cervical cancer yet due to their young age), yet the authors seem to imply these can be effective in the Discussion ("feasible and can yield desirable outcomes"). Can the authors comment on what a clinically relevant difference in time to diagnosis is for breast and cervical cancer?
Response: We appreciate this insightful feedback. That reduction in time to diagnosis was from a breast cancer education programme targeted at women (not school girls). We have discussed the result more cautiously (Lines 406 – 416)

Comment: Most of the studies do not include measures for time to diagnosis. I realize this is one of the points the authors are trying to make – that too few studies consider timeliness. Is there a way to provide a summary / number for timeliness, since it is a main outcome of interest in the paper?
Response: Thank you for this. As acknowledged in the discussion (lines 446 – 447), the highly heterogeneous nature of the studies and their contextual settings undermined the quantitative aggregation and comparability of findings across the few number of studies that reported time-related outcomes.

VERSION 2 – REVIEW

REVIEWER	Lydia Pace Brigham and Women's Hospital, Boston, Massachusetts, Medicine
REVIEW RETURNED	06-Mar-2022

GENERAL COMMENTS	Thank you for the opportunity to re-review this work. The authors have responded adequately to my comments and the manuscript has improved significantly, particularly the discussion which is very well written and makes important points about both research and implementation needs for early diagnosis. 2 comments: -for future work, in Response to Review documents, it is very helpful for authors to directly quote the edits made, within the Response document (rather than just noting the relevant lines of the paper), so that the reviewer can quickly link the initial feedback to the edits. -The Pace et al RCT – the authors describe this as an intervention targeting CHWs but the intervention included: training for community health workers, training for health center nurses, training for hospital level clinicians in diagnostic breast ultrasound, and establishment of weekly breast clinics at health centers and the hospital (see "Methods – Intervention" in that paper) – not just a CHW intervention. It is appropriately included as an intervention targeting Step 2 in the table despite the intervention being somewhat mischaracterized, but since the intervention included step 1 and step 2, the authors may wish to include it as an "Intervention targeting step 1 and step 2." I am admittedly less intimately familiar with some of the other studies described so it is possible that there are other misclassifications and it may be worth a quick look to make sure that the interventions are accurately described.

VERSION 2 – AUTHOR RESPONSE

Reviewer 1:

Comment: Thank you for the opportunity to re-review this work. The authors have responded adequately to my comments and the manuscript has improved significantly, particularly the discussion

which is very well written and makes important points about both research and implementation needs for early diagnosis.
Response: We appreciate this feedback and are again grateful for the previous comments, which we found really helpful for the revisions made.

Comment: For future work, in Response to Review documents, it is very helpful for authors to directly quote the edits made, within the Response document (rather than just noting the relevant lines of the paper), so that the reviewer can quickly link the initial feedback to the edits.
Response: This is noted. Thank you.

Comment: The Pace et al RCT – the authors describe this as an intervention targeting CHWs but the intervention included: training for community health workers, training for health center nurses, training for hospital level clinicians in diagnostic breast ultrasound, and establishment of weekly breast clinics at health centers and the hospital (see "Methods – Intervention" in that paper) – not just a CHW intervention. It is appropriately included as an intervention targeting Step 2 in the table despite the intervention being somewhat mischaracterized, but since the intervention included step 1 and step 2, the authors may wish to include it as an "Intervention targeting step 1 and step 2." I am admittedly less intimately familiar with some of the other studies described so it is possible that there are other misclassifications and it may be worth a quick look to make sure that the interventions are accurately described.
Response: We appreciate this insightful feedback and agree that the intervention (though primarily targeting step 2) also targets step 1 through CHW community education. We have therefore moved it to fall under interventions targeting steps 1 and 2 in the table. We have also clarified the description of the intervention: *"It consisted of training of CHWs in how to educate community members about breast cancer symptoms recognition and early help-seeking; training of nurses in symptom recognition, CBE and appropriate referral; training of hospital-based clinicians in diagnostic breast ultrasound and ultrasound-guided core needle biopsy"*. (See Table 1). We confirm that all other interventions are characterised appropriately.